

Markov Transfer Operators for Covariance-Matrix Dynamics

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Abstract

This note records a precise formulation of the stochastic Gaussian covariance update as a discrete-time Markov process on covariance matrices. Each elementary circuit event is a state-dependent random rational map $G \mapsto F_\alpha(G)$, where the branch label α contains the local operation and measurement outcome. The corresponding Markov kernel pushes forward probability distributions on the covariance manifold, while its adjoint acts on observables by the usual Koopman transfer operator. Linearizing the same random maps gives a multiplicative tangent cocycle $\delta G_{t+1} = T_{\alpha_t}(G_t)^\dagger \delta G_t T_{\alpha_t}(G_t)$. The Lyapunov spectrum is the quenched singular-value spectrum of this cocycle, not the spectrum of an outcome-averaged transfer matrix.

I. COVARIANCE STATE SPACE

Let $G \in \mathbb{C}^{N \times N}$ denote the complex covariance matrix of the top-layer Gaussian state. In the convention used here,

$$G_{ij} = 2\langle c_i^\dagger c_j \rangle - \delta_{ij}. \quad (1)$$

Thus the one-body occupation matrix is

$$C = \frac{1}{2}(G + \mathbb{I}). \quad (2)$$

The natural state space is a real manifold, or a stratified real analytic set, of admissible covariance matrices,

$$\mathcal{X} = \{G : G = G^\dagger, -\mathbb{I} \leq G \leq \mathbb{I}, \text{ plus the imposed symmetry constraints}\}. \quad (3)$$

Pure Gaussian states lie on the boundary component satisfying

$$G^2 = \mathbb{I}. \quad (4)$$

A perturbation away from a pure state obeys

$$(G + \delta G)^2 = \mathbb{I} + \{G, \delta G\} + O(\delta G^2). \quad (5)$$

Therefore a first-order tangent perturbation along the pure-state manifold satisfies $\{G, \delta G\} = 0$, whereas first-order mixedness is encoded by the normal component for which $\{G, \delta G\} \neq 0$.

II. ELEMENTARY BRANCH MAPS

An elementary realized Gaussian channel branch acts on G by a fractional linear covariance map

$$F_\alpha(G) = C_\alpha - B_\alpha^\dagger (A_\alpha - G^{-1})^{-1} B_\alpha, \quad (6)$$

where α labels the operation, site or mode, direction, and measurement outcome. The domain of this formula is the open subset of $\mathcal{X}$ on which the inverse exists. Singular events should be understood either by restricting to the nonsingular support block or by taking the appropriate Schur-complement limit.

For a filling or deletion operation at mode (r, ν) , a useful branch label is

$$\alpha = (r, \nu, \sigma, s), \quad \sigma \in \{\text{fill}, \text{del}\}, \quad s \in \{0, 1\}. \quad (7)$$

Let

$$P_{r,\nu} = W_{r,\nu} W_{r,\nu}^\dagger, \quad Q_{r,\nu} = \mathbb{I} - P_{r,\nu}. \quad (8)$$

The branch probability is state dependent. For instance, for an occupation measurement of the mode $W_{r,\nu}$,

$$p_{r,\nu}(s = 1 \mid G) = \text{Tr}\left(P_{r,\nu} \frac{G + \mathbb{I}}{2}\right), \quad p_{r,\nu}(s = 0 \mid G) = 1 - p_{r,\nu}(s = 1 \mid G), \quad (9)$$

up to the precise sign convention used to associate the two branch labels with occupied and empty outcomes. More general schedules may also include an externally chosen probability for selecting (r, ν, σ) .

III. DISCRETE-TIME MARKOV PROCESS ON COVARIANCE MATRICES

The covariance process is a time-inhomogeneous Markov chain on $\mathcal{X}$. Given the current covariance G_t , one samples a branch label α_t from a conditional probability law

$$\mathbb{P}(\alpha_t = \alpha \mid G_t = G) = p_t(\alpha \mid G), \quad (10)$$

and then updates deterministically inside that branch,

$$G_{t+1} = F_\alpha(G_t). \quad (11)$$

Equivalently,

$$G_{t+1} = F_{\alpha_t}(G_t), \quad \alpha_t \sim p_t(\cdot \mid G_t). \quad (12)$$

The Markov property is immediate:

$$\mathbb{P}(G_{t+1} \in E \mid G_t, G_{t-1}, \dots, G_0) = \mathbb{P}(G_{t+1} \in E \mid G_t) \quad (13)$$

for every Borel set $E \subset \mathcal{X}$, provided the circuit schedule at time t is fixed or is itself included as part of the time-dependent law p_t .

The associated Markov kernel is

$$K_t(G, E) = \sum_{\alpha \in \mathcal{A}_t} p_t(\alpha \mid G) \mathbf{1}_E(F_\alpha(G)), \quad (14)$$

or, in distributional notation,

$$K_t(G, dG') = \sum_{\alpha \in \mathcal{A}_t} p_t(\alpha \mid G) \delta_{F_\alpha(G)}(dG'). \quad (15)$$

Here $\mathcal{A}_t$ is the finite set of branch labels available at step t . For a deterministic choice of location and operation, $\mathcal{A}_t$ contains only the two measurement outcomes. For a randomly selected location or operation, $\mathcal{A}_t$ contains the combined operation-outcome labels and p_t includes both the external sampling probability and the Born probability.

IV. TRANSFER OPERATORS ON OBSERVABLES AND DISTRIBUTIONS

Let $\varphi : \mathcal{X} \rightarrow \mathbb{C}$ be a bounded measurable observable of the covariance matrix. The Koopman, or backward, Markov transfer operator is

$$(\mathfrak{K}_t \varphi)(G) = \mathbb{E}[\varphi(G_{t+1}) \mid G_t = G] = \sum_{\alpha \in \mathcal{A}_t} p_t(\alpha \mid G) \varphi(F_\alpha(G)). \quad (16)$$

The dual Perron-Frobenius operator evolves probability measures. If μ_t is the law of G_t , then

$$\mu_{t+1} = \mathfrak{P}_t \mu_t, \quad (\mathfrak{P}_t \mu_t)(E) = \int_{\mathcal{X}} K_t(G, E) \mu_t(dG). \quad (17)$$

Equivalently, duality with observables gives

$$\int_{\mathcal{X}} \varphi(G') \mu_{t+1}(dG') = \int_{\mathcal{X}} (\mathfrak{K}_t \varphi)(G) \mu_t(dG). \quad (18)$$

If μ_t has a density ρ_t with respect to a chosen reference measure dG on the real covariance manifold, then formally

$$\rho_{t+1}(G') = \sum_{\alpha \in \mathcal{A}_t} \int_{\mathcal{X}} \rho_t(G) p_t(\alpha \mid G) \delta(G' - F_\alpha(G)) dG. \quad (19)$$

When each branch map is locally invertible on the relevant smooth stratum, this can be written as

$$\rho_{t+1}(G') = \sum_{\alpha \in \mathcal{A}_t} \sum_{G \in F_\alpha^{-1}(G')} \frac{\rho_t(G) p_t(\alpha \mid G)}{|\det_{\mathbb{R}} DF_\alpha(G)|}. \quad (20)$$

The determinant in Eq. (20) is the real Jacobian on the appropriate tangent space of admissible covariance matrices. In practice, Eq. (17) is the more robust definition: measurement maps can be rank deficient or singular on lower-dimensional sets, in which case the evolved measure need not remain absolutely continuous.

V. FULL LAYERS AND UNIT CELLS

A full circuit layer or unit cell consists of a finite sequence of elementary updates. Suppose the physical order is

$$j = 1, 2, \dots, m, \quad (21)$$

with elementary kernels $K_t^{(j)}$. The covariance process inside the cell is

$$G^{(j)} = F_{\alpha^{(j)}}^{(j)}(G^{(j-1)}), \quad \alpha^{(j)} \sim p_t^{(j)}(\cdot \mid G^{(j-1)}). \quad (22)$$

The cell-level kernel is the kernel composition

$$K_t^{\text{cell}} = K_t^{(1)} K_t^{(2)} \dots K_t^{(m)} \quad (23)$$

in the following explicit sense:

$$K_t^{\text{cell}}(G, E) = \int_{\mathcal{X}} \dots \int_{\mathcal{X}} K_t^{(1)}(G, dG^{(1)}) K_t^{(2)}(G^{(1)}, dG^{(2)}) \dots K_t^{(m)}(G^{(m-1)}, E). \quad (24)$$

Thus for observables,

$$\mathfrak{K}_t^{\text{cell}} \varphi = \mathfrak{K}_t^{(1)} \mathfrak{K}_t^{(2)} \dots \mathfrak{K}_t^{(m)} \varphi, \quad (25)$$

where the first physical operation appears first because $(\mathfrak{K}_t^{(1)} \psi)(G) = \mathbb{E}[\psi(G^{(1)}) \mid G^{(0)} = G]$.

For measures the dual order is

$$\mu_t^{\text{out}} = \mathfrak{P}_t^{(m)} \dots \mathfrak{P}_t^{(2)} \mathfrak{P}_t^{(1)} \mu_t^{\text{in}}. \quad (26)$$

VI. LINEARIZATION AND TANGENT TRANSFER MATRICES

The branchwise tangent map follows by differentiating Eq. (6). Let

$$M_\alpha(G) = A_\alpha - G^{-1}. \quad (27)$$

For a perturbation δG ,

$$\delta G^{-1} = -G^{-1} \delta G G^{-1}, \quad (28)$$

and therefore

$$\delta M_\alpha = G^{-1} \delta G G^{-1}. \quad (29)$$

Since $\delta M^{-1} = -M^{-1}(\delta M)M^{-1}$, one obtains

$$DF_\alpha(G)[\delta G] = B_\alpha^\dagger M_\alpha(G)^{-1} G^{-1} \delta G G^{-1} M_\alpha(G)^{-1} B_\alpha. \quad (30)$$

Using the noncommutative identities

$$M_\alpha(G)^{-1} G^{-1} = (GA_\alpha - \mathbb{I})^{-1}, \quad G^{-1} M_\alpha(G)^{-1} = (A_\alpha G - \mathbb{I})^{-1}, \quad (31)$$

this becomes

$$DF_\alpha(G)[\delta G] = B_\alpha^\dagger (GA_\alpha - \mathbb{I})^{-1} \delta G (A_\alpha G - \mathbb{I})^{-1} B_\alpha. \quad (32)$$

When the covariance and Choi blocks obey the Hermiticity relation needed to identify the left factor as the adjoint of the right factor, define

$$T_\alpha(G) = (A_\alpha G - \mathbb{I})^{-1} B_\alpha. \quad (33)$$

Then

$$\delta G_{t+1} = T_{\alpha_t}(G_t)^\dagger \delta G_t T_{\alpha_t}(G_t). \quad (34)$$

VII. LIFTED MARKOV PROCESS AND LYAPUNOV COCYCLE

Equations (11) and (34) define a Markov process lifted to the tangent bundle:

$$G_{t+1} = F_{\alpha_t}(G_t), \quad (35)$$

$$\delta G_{t+1} = T_{\alpha_t}(G_t)^\dagger \delta G_t T_{\alpha_t}(G_t), \quad (36)$$

$$\alpha_t \sim p_t(\cdot \mid G_t). \quad (37)$$

Along one realized trajectory, set

$$T_t := T_{\alpha_t}(G_{t-1}). \quad (38)$$

Then the chronological product convention compatible with $\delta G_t = T_t^\dagger \delta G_{t-1} T_t$ is

$$\tilde{T}_n = T_1 T_2 \cdots T_n, \quad (39)$$

so that

$$\delta G_n = \tilde{T}_n^\dagger \delta G_0 \tilde{T}_n. \quad (40)$$

The Lyapunov exponents are the quenched singular-value growth rates

$$\lambda_i = \lim_{n \rightarrow \infty} \frac{1}{n} \log \sigma_i(\tilde{T}_n), \quad (41)$$

when the limit exists. Equivalently,

$$2\lambda_i = \lim_{n \rightarrow \infty} \frac{1}{n} \log \text{eig}_i \left(\tilde{T}_n^\dagger \tilde{T}_n \right). \quad (42)$$

In a stationary ergodic setting, existence and nonrandomness of the limiting exponents are the content of the multiplicative ergodic theorem applied to the Markov cocycle $T_\alpha(G)$.

VIII. BRANCH AVERAGING VERSUS QUENCHED PRODUCTS

A useful distinction is between the annealed distributional dynamics of G and the quenched tangent dynamics along a realized trajectory. The distribution of G evolves by the Markov kernel,

$$\mu_{t+1} = \mathfrak{P}_t \mu_t. \quad (43)$$

However, the tangent update is quadratic in the branch transfer matrix. Thus, conditional on $G_t = G$,

$$\mathbb{E}[\delta G_{t+1} \mid G_t = G, \delta G_t] = \sum_{\alpha \in \mathcal{A}_t} p_t(\alpha \mid G) T_\alpha(G)^\dagger \delta G_t T_\alpha(G). \quad (44)$$

This is not equal in general to

$$\bar{T}(G)^\dagger \delta G_t \bar{T}(G), \quad \bar{T}(G) = \sum_{\alpha \in \mathcal{A}_t} p_t(\alpha \mid G) T_\alpha(G). \quad (45)$$

Consequently the Lyapunov spectrum relevant to relaxation of perturbations is the spectrum of the random product in Eq. (39), not the spectrum of an outcome-averaged matrix.

IX. SUMMARY

The covariance dynamics can be organized at three levels. First, the state G_t is a discrete-time Markov chain with transition kernel $K_t(G, dG')$. Second, distributions of G_t evolve by the Perron-Frobenius pushforward $\mathfrak{P}_t$, while observables evolve by the dual Koopman operator $\mathfrak{K}_t$. Third, infinitesimal perturbations define a matrix cocycle over the Markov chain,

$$(G_t, \delta G_t) \mapsto (F_{\alpha_t}(G_t), T_{\alpha_t}(G_t)^\dagger \delta G_t T_{\alpha_t}(G_t)) . \quad (46)$$

The Lyapunov spectrum is obtained from the singular values of the chronological product of the realized tangent transfer matrices.